

6. MANUAL CALCULATION OF NEWTON-RAPHSON LOAD FLOW

GIVEN DATA

Bus Types

Bus 1 : Slack Bus

Bus 2 : PV Bus

Bus 3 : PQ Bus

Specified Powers

$$P_2 = -4.0 \text{ p.u.}$$

$$P_3 = -6.0 \text{ p.u.}$$

$$Q_3 = -4.0 \text{ p.u.}$$

Initial Voltages

$$V_1 = 1.01 \angle 0^\circ$$

$$V_2 = 1.03 \angle 0^\circ$$

$$V_3 = 1.00 \angle 0^\circ$$

$$\text{Tolerance} = 10^{-6}$$

$$\text{Maximum number of iterations} = 10$$

SOLUTION

STEP 1 : CALCULATION OF LINE ADMITTANCES

$$Y_{12} = -j5$$

$$Y_{23} = -1/(0.02+j0.04) - 1/(0.05+j0.025)$$

$$= (-10+j20) + (-16+j8)$$

$$= -26+j28$$

$$Y_{13} = 0$$

STEP 2 : FORMATION OF Y-BUS MATRIX

$$Y_{11} = j5$$

$$Y_{22} = 26-j23$$

$$Y_{33} = 26-j28$$

$$Y_{12} = Y_{21} = -j5$$

$$Y_{23} = Y_{32} = -26+j28$$

$$Y_{13} = Y_{31} = 0$$

Hence,

$$Y_{BUS} =$$

$$\begin{bmatrix} j5 & -j5 & 0 \\ -j5 & 26-j23 & -26+j28 \\ 0 & -26+j28 & 26-j28 \end{bmatrix}$$

STEP 3 : INITIAL VALUES

$$V_1 = 1.01 \angle 0^\circ$$

$$V_2 = 1.03 \angle 0^\circ$$

$$V_3 = 1.00 \angle 0^\circ$$

$$\delta_1 = 0^\circ$$

$$\delta_2 = 0^\circ$$

$$\delta_3 = 0^\circ$$

STEP 4 : POWER MISMATCH CALCULATION

Using

$$\Delta P = P_{\text{specified}} - P_{\text{calculated}}$$

$$\Delta Q = Q_{\text{specified}} - Q_{\text{calculated}}$$

Mismatch vector

$$\Delta X = [\Delta P_2 \ \Delta P_3 \ \Delta Q_3]^T$$

STEP 5 : JACOBIAN MATRIX FORMATION

The Jacobian matrix is formed as

$$J = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$

The correction vector is obtained from

$$J \Delta X = \text{Mismatch}$$

The updated values are

$$\delta_{\text{new}} = \delta_{\text{old}} + \Delta\delta$$

$$V_{\text{new}} = V_{\text{old}} + \Delta V$$

STEP 6 : ITERATIONS

The process is repeated until

$$\text{Maximum mismatch} < 10^{-6}$$

The solution converges after 10 iterations.

FINAL BUS VOLTAGES

Bus 1

$$V = 1.0100 \text{ p.u.}$$

$$\text{Angle} = 0.0000^\circ$$

Bus 2

$$V = 1.0300 \text{ p.u.}$$

$$\text{Angle} = 1525.8812^\circ$$

Bus 3

$$V = 0.0924 \text{ p.u.}$$

$$\text{Angle} = 1574.3123^\circ$$

$$\text{Bus 1 : } V = 1.0100 \text{ p.u., Angle} = 0.0000^\circ$$

$$\text{Bus 2 : } V = 1.0300 \text{ p.u., Angle} = 1525.8812^\circ$$

$$\text{Bus 3 : } V = 0.0924 \text{ p.u., Angle} = 1574.3123^\circ$$